

### Open RAN components

- **Baseband unit:** This transports a baseband frequency, usually from a remote radio unit, to which it may be tied through optical fibre.
- **Remote radio unit:** RRU is the distributed and integrated frequency unit that connects to an operator's network with the user equipment (like cell phone and mobile devices).
- **Radio frequency:** RF band is a frequency band that is used for transmissions and broadcasting.
- **Common public radio interface (CPRI):** CPRI is a popular standard for transporting baseband signals to radio units. It is an interface between radio equipment control (REC) and radio equipment (RE).

V1 with a new vendor V2. Because of the open interface, the hardware from vendor A, B or C will work well with the software from vendor V2 too.

### What Open RAN enables

- **No vendor lock-in:** With the fully interoperable and desegregated RAN, the service provider is able to choose from the RAN component manufacturers. This results in a mix of vendors as opposed to just one or two.
- **Best-of-breed components:** Open RAN brings more flexibility in choosing components from the best vendors. It promotes supply chain diversification as well as price competition among vendors.
- **Reduced costs:** Open RAN helps to reduce costs by enabling rapid innovation, vendor diversification and higher automation.

### Challenges with Open RAN

- **Security:** Network security is one of the major challenges as different vendors follow different security protocols.
- **Openness:** Since it is open source, it is susceptible to various kinds of threats.

Mobile network operators are benefiting tremendously with Open RAN standards. New features on the anvil will make O-RAN even more economical, while improving hardware and software integration. **END** 🐧

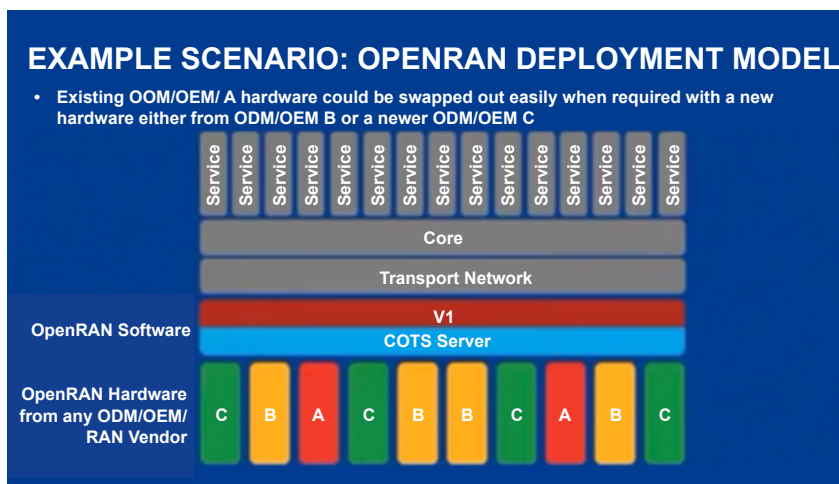


Figure 5: Open RAN deployment model

purchased from any original design manufacturer (ODM), original equipment manufacturer (OEM) or RAN hardware vendor. The BBU is proprietary software with virtualised functions running on a COTS server. It is important to note that the interface between BBU and RRU is an open interface, as can be seen in Figure 4. This will allow any vendor's software to work with a different vendor's hardware.

The example shown in Figure 5 will help to understand this better. Let us assume that a service provider has deployed O-RAN software from vendor V1 that runs on a COTS server, while the hardware is being used from an OEM/ODM vendor A. If the service provider now decides to take components from a new hardware vendor, vendor B or

vendor C (maybe because of better features), this hardware will also be compatible with the O-RAN software from vendor V1. Now, let us say that the service provider is unhappy with the software vendor V1 for some reason. In that case, he can just replace vendor

### References

- [1] <https://www.youtube.com/user/ParallelWireless>
- [2] <https://www.parallelwireless.com/blog/dispelling-the-fear-uncertainty-and-doubt-about-open-ran-integration/>

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